

制造业案例中科类脑 基于Volcano为电力设备装上智能大脑

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01

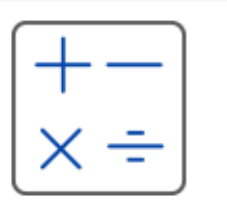
Leinao YunOS 介绍



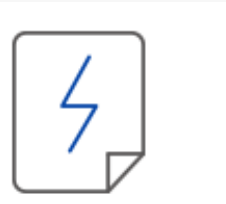
Leinao OS 介绍



提供高性能计算集群提供多种CPU/GPU卡



算法开发一键部署，全栈开发



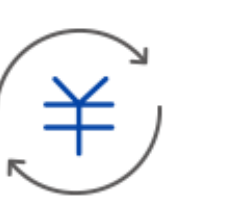
快速部署1个节点，100个节点，部署同样轻松



核心技术最安心的硬核系统



界面友好直接上手



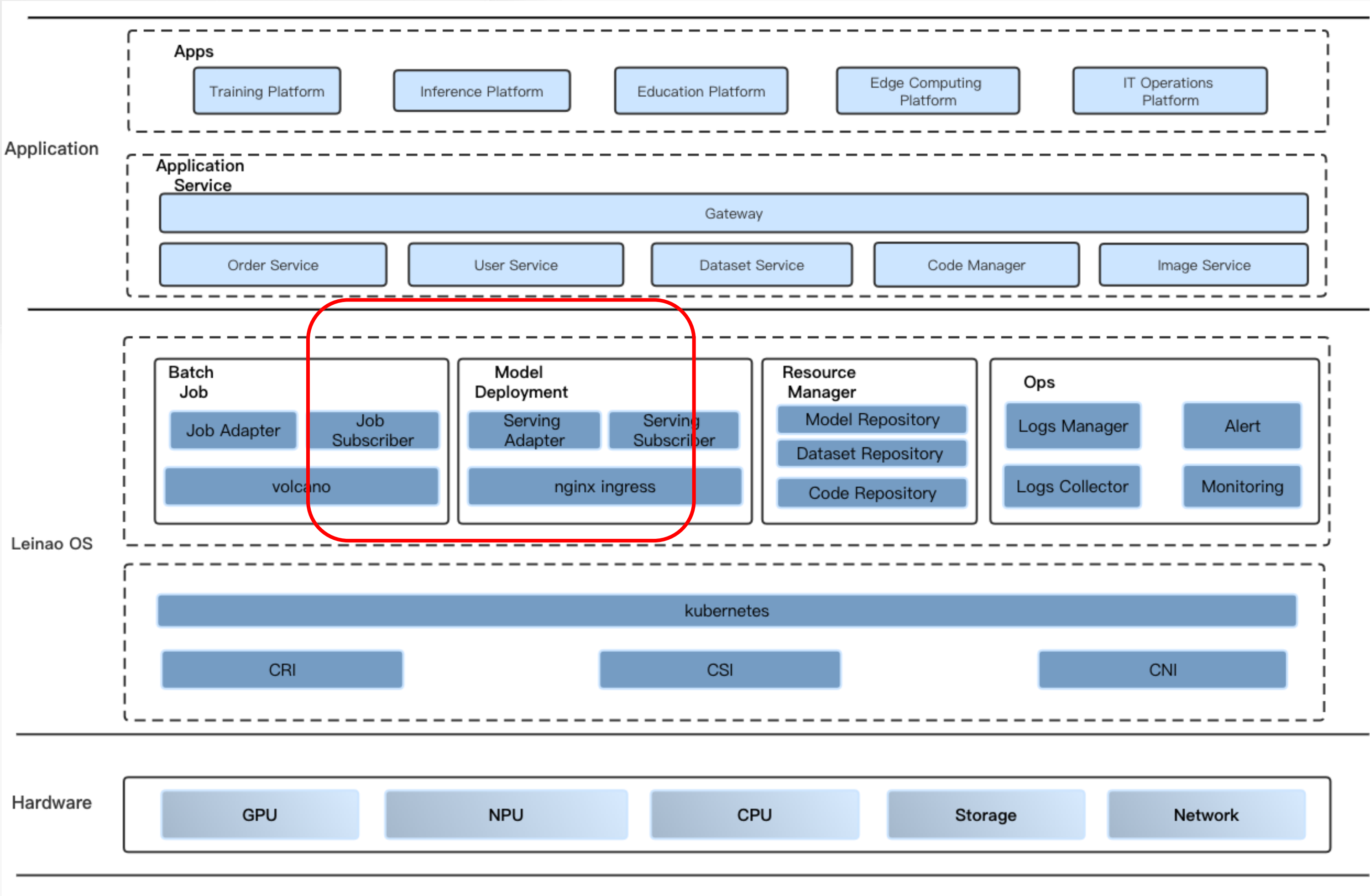
资产管理内置GIT，安全保护你的IP

“类脑智能开放平台”由类脑智能技术与应用国家工程实验室和中国科技类脑公司共同建设，拥有自己的高性能计算集群。该平台自2019年10月起全面向科研机构、大学和个人AI开发者开放，为用户提供高性能计算资源，为快速构建和训练模型提供环境，并不断更新开箱即用的选定算法和数据集，让开发者专注于商业和科学研究。



基于k8s，围绕模型，以模型训练为核心，同时支持多种业务产品。包括数据处理、数据集管理、编码调试、模型训练、模型管理和模型部署等所有操作。

Leinao YunOS 系统架构



Leinao YunOS System Architecture Diagram

02

批处理作业定义



为什么选择Volcano Job CRDs

CRD	应用场景	不匹配原因
Deployment	Stateless service	Job type mismatch
DaemonSet	All matching nodes must deploy one pod	Deployment scenario not applicable
StatefulSet	Stateful service	Job lifecycle mismatch
Job	Simple batch job	Does not support complex distributed training

AI批处理作业特性:

1. 需要重试机制

2. 需要优先级和作业队列设置

3. 需要选择特定的作业和节点亲和性

4. 需要节点间IP感知

5. 需要节点间通信保证和网络隔离
6. 需要某些启动顺序依赖

7. 部分场景需要自动配置SSH免密码登录

8. 需要最小值来满足资源设置

9. 需要资源池划分和指定

10. 需要作业故障转移功能



批处理作业定义-01

- Queue
- Job And Task Retry Strategy
- SSH Plugin

sample volcano job

```
apiVersion: batch.volcano.sh/v1alpha1
kind: Job
metadata:
  name: sample-job
spec:
  maxRetry: 3
  minAvailable: 2
  plugins:
    ssh: []
  queue: default
  schedulerName: volcano
  priorityClassName: high-priority
tasks:
  - maxRetry: 3
    minAvailable: 0
    name: task1
    replicas: 1
    .....
```

volcano queue example

```
apiVersion: scheduling.volcano.sh/v1beta1
kind: Queue
metadata:
  name: default
spec:
  reclaimable: true
  weight: 1
status:
  allocatable:
    cpu: "128"
    intel.com/infiniband_rdma_netdevs: "4"
    memory: 535545451861m
    nvidia.com/gtx-1080-ti: "8"
  inqueue: 1
  running: 10
  state: Open
```

run new task

custom configuration

Cancel custom configuration

* mirror

pytorch1.6_mmdetection

* queue

default

* GPU type

gtx-1080-ti

Task failed restart

3

task1

task2

+ add task

* subtask name

task1

* number of subtasks

2

* Number of GPUs

1

Minimum number of successes

1

Minimum number of failures

1

* Minimum number of runs

1

* start command

python train.py

* payment method

Pay with personal account

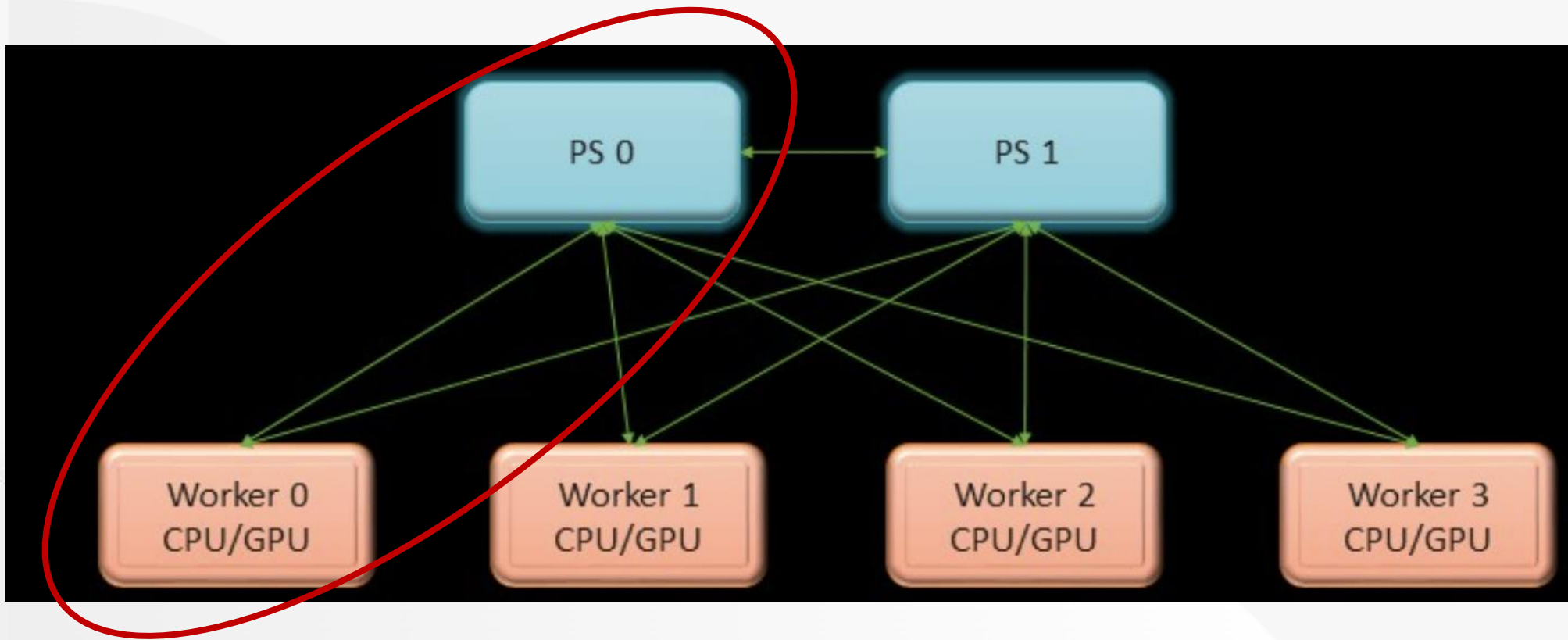
remaining 687.73 computing power

Selected configuration: 1 gtx-1080-ti card, 2-core CPU, 16G memory, 8192M shared memory

Create Batch Job In Leinao Yun OS

批处理作业定义-02

- Job and Task Minavailable
- IP Awareness
- Task Restart Policies
- Job Network Policy
- Job startup dependencies



TensorFlow distributed training example

The minimum number of replicas required for a job to start

To support Job communication through K8s Service

Minimum number of replicas required for job startup

Maximum number of copies of a job

Policies corresponding to different states of tasks

Startup order dependencies between tasks

```
# tensorflow distribute job example
apiVersion: batch.volcano.sh/v1alpha1
kind: Job
metadata:
  name: tensorflow-distribute-job
spec:
  maxRetry: 1
  minAvailable: 2
  plugins:
    svc: []
  schedulerName: volcano
  tasks:
    - maxRetry: 1
      minAvailable: 1
      name: ps
      replicas: 2
      policies:
        - action: RestartJob
          event: PodFailed
        - action: RestartJob
          event: PodEvicted
        - action: CompleteJob
          event: TaskCompleted
      .....
    - maxRetry: 1
      minAvailable: 1
      name: worker
      replicas: 4
      dependsOn:
        name: ps
      .....

```

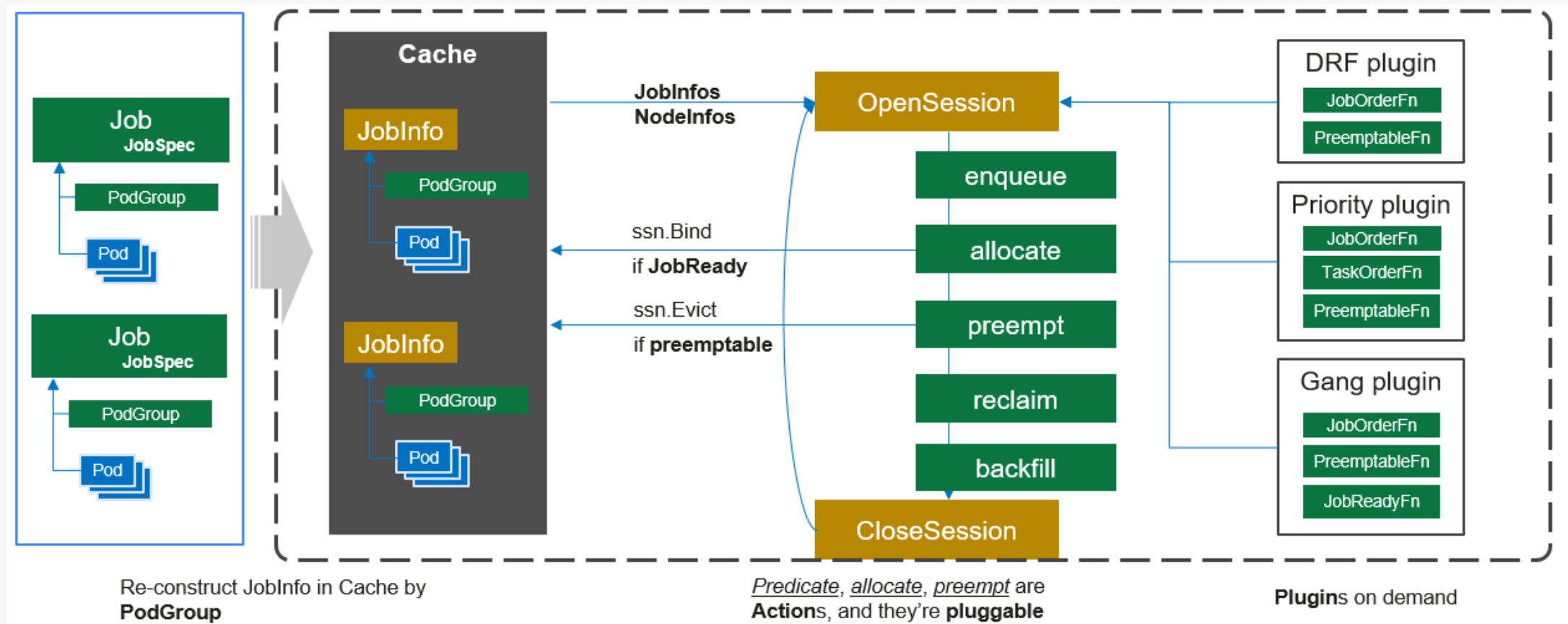
03

批处理作业调度



为什么选择Volcano Scheduler

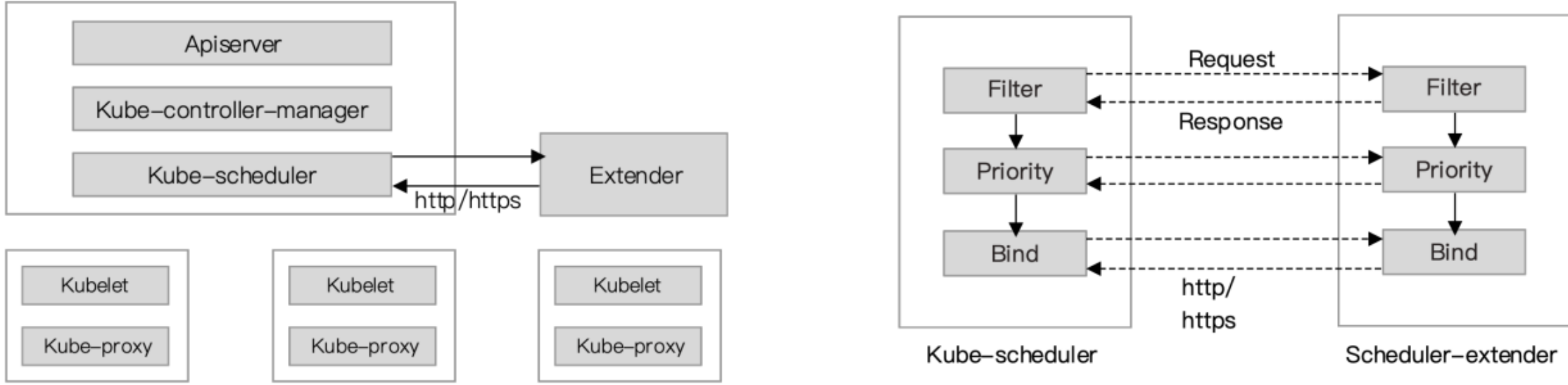
Volcano Scheduler



- 支持多种调度算法
- 更高效的调度
- 非侵入式支持主流计算框架

Kube Scheduler

Scheduler Extender



- 通过HTTP请求调用扩展器接口，性能远低于本地函数调用。
- 用户可扩展的点数有限，位置相对固定，无法支持灵活扩展。
- 对批处理作业调度场景的支持较少。

AI训练作业对于调度器的要求:

1. 支持为集群划分资源池
2. 支持整体作业调度
3. 支持节点资源预留
4. 支持按比例划分节点资源
5. 支持按优先级调度任务
6. 支持节点和作业之间的亲和性调度
7. 支持基本资源检查和调度条件检查
8. 支持通过水平弹性伸缩来加速训练

批处理作业调度-01

Proportional Plugin

Node

node	CPU	Memory	Nvidia/GTX-1080-TI
node-A	40	100	8
node-B	40	100	0

Task

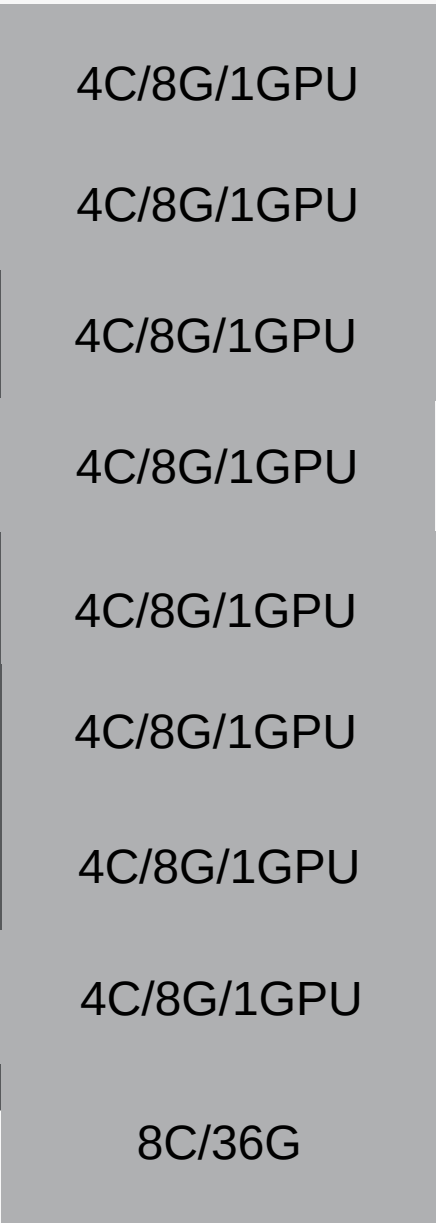
Task	CPU	Memory	Nvidia/GTX-1080-TI	Bind
Task-1	40	1	1	node-A
Task-2	4	4	1	Pending

result of unproportionalscheduling

Full node
resource



Proportionately
divide node
resources



volcano-scheduler.conf

actions: "enqueue, allocate, backfill"

tiers:

- plugins:
- name: priority
- name: gang
- name: conformance
- name: predicates

arguments:

predicate.ProportionalEnable: true

predicate.resources: nvidia.com/gtx-1080-ti

predicate.resources.nvidia.com/gtx-1080-ti.cpu: 4

predicate.resources.nvidia.com/gtx-1080-ti.memory: 8

Task

Task	CPU	Memory	Nvidia/GTX-1080-TI	Bind
Task-1	40	1	0	node-B
Task-2	4	4	1	node-A
Task-3	10	10	0	Pending

result of proportional scheduling

Conditions:

$\text{node.Idel.CPU} - \text{task.Resreq.CPU} < \text{node.Idel.GPU} * \text{cpuRatio} \parallel$

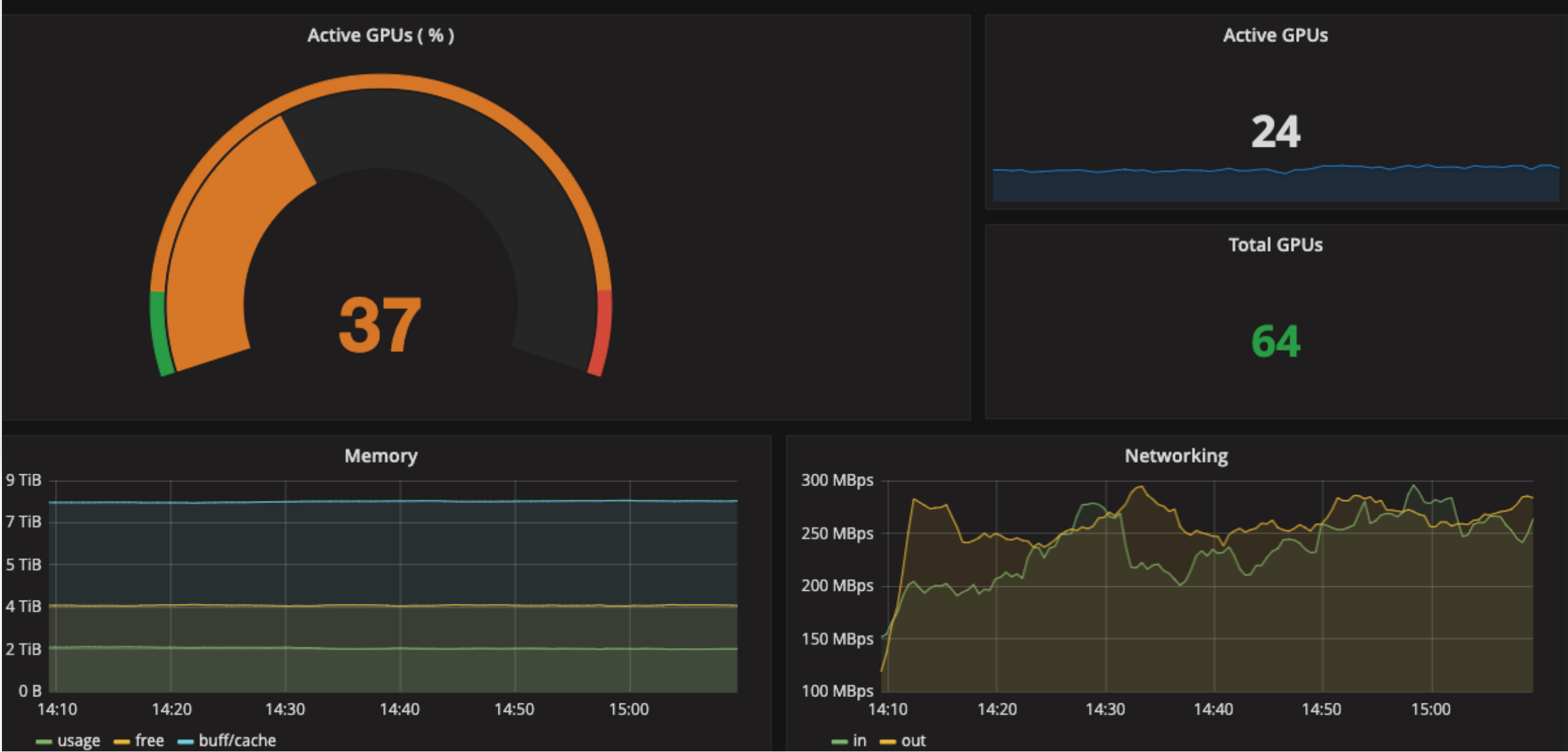
$\text{node.Idel.Memory} - \text{task.Resreq.Memory} < \text{node.Idel.GPU} * \text{memoryRatio}$

Resource Fragmentation

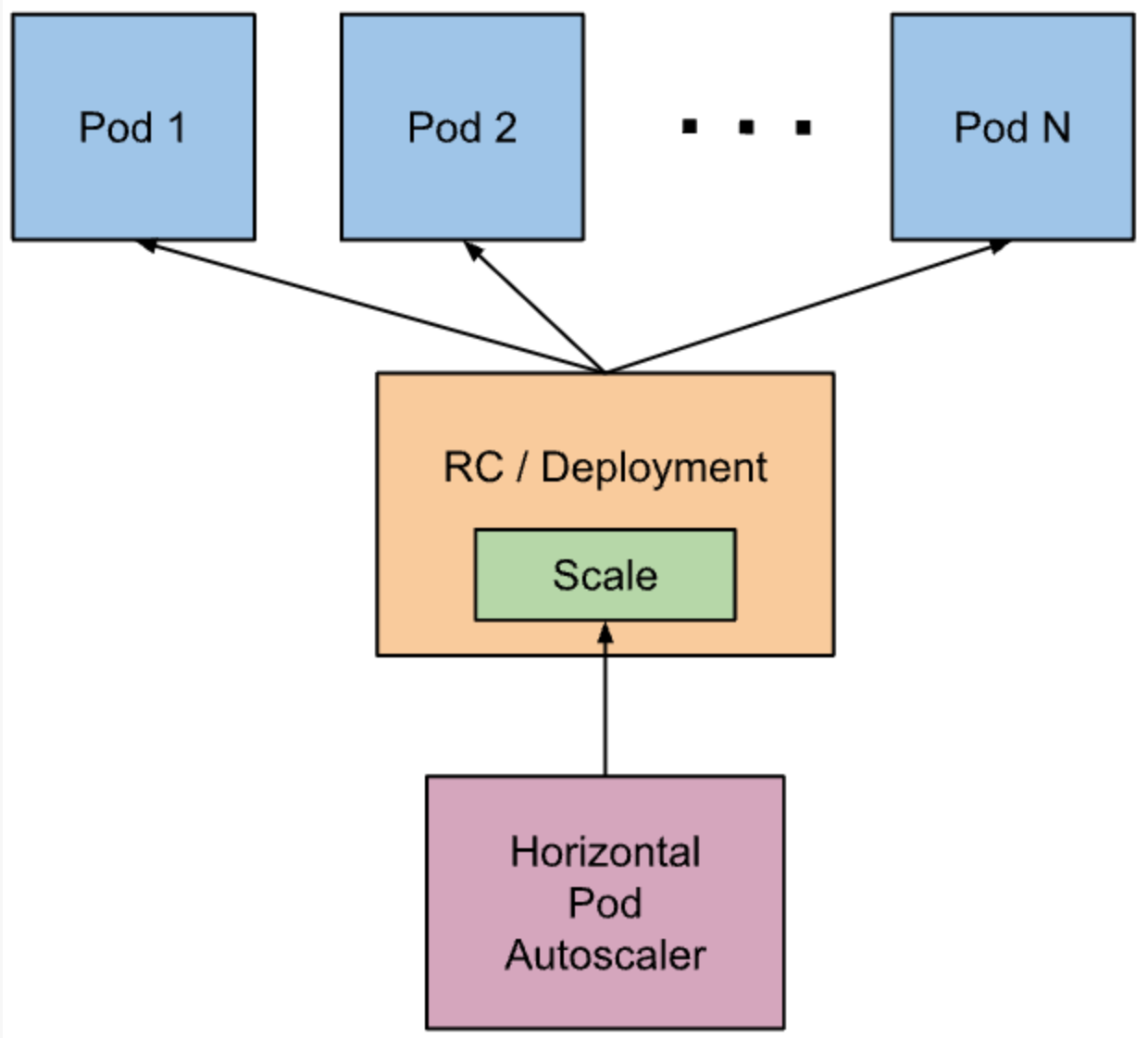


批处理作业调度-02

Elastic Distribute Training



Cluster resource utilization

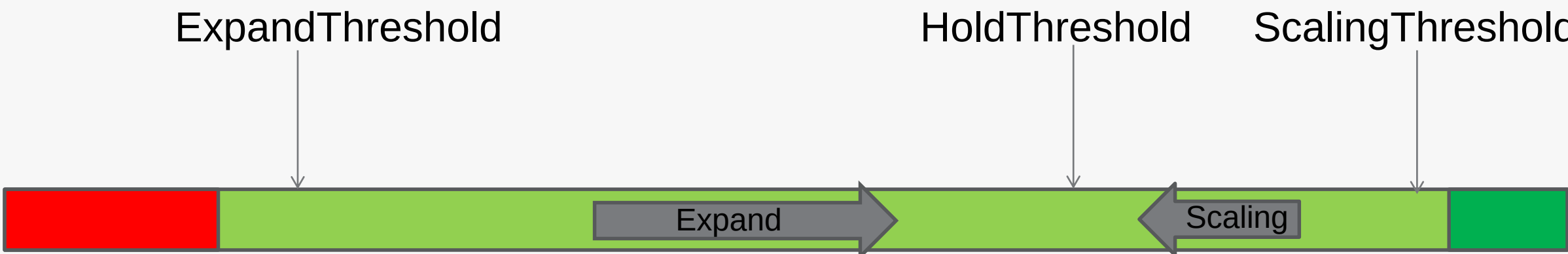


Kubernetes Horizontal Pod Autoscaler

$$\text{desiredReplicas} = \text{ceil} \left[\frac{\text{currentReplicas} * \text{currentMetricValue}}{\text{desiredMetricValue}} \right]$$



Schematic diagram of resource division of elastic training cluster



Schematic diagram of elastic scaling



Horovod Elastic Training Adapter

example

```
horovodrun -np 2 --min-np 2 --max-np 5 \
  --host-discovery-script discover_hosts.sh \
  python tensorflow2_synthetic_benchmark_elastic.py
```

- **np**
The elastic training job will not start until at least -np slots are available for running worker processes.
- **min-np**
Below --min-np (due to host failure, preemption, etc.), then the job will pause waiting for more hosts to become available or until HOROVOD_ELASTIC_TIMEOUT (default: 600 seconds) has elapsed.
- **max-np**
The max np number.
- **host-discovery-script**
Horovod to discover available hosts script.

host discovery script output example

```
horovod-elastic-job-master-0.horovod-elastic-job:1
horovod-elastic-job-worker-0.horovod-elastic-job:1
...when pod replicas increase....
horovod-elastic-job-worker-3.horovod-elastic-job:1
```

run new task

Recommended configuration

custom configuration

* mirror

pytorch1.6_mmdetection

* queue

default

* GPU type

gtx-1080-ti

* Number of C

1

* start command

python tensorflow2_synthetic_benchmark_elastic.py

☒ Enable elastic scaling

Turning on elastic scaling will automatically speed up training for you when the platform resources are

Idle, learn more

* number

2

* maximum np

5

Create Elastic Distributed Training Job in Leinao YunOS

```
spec:
  plugins:
    svc: []
    ssh: []
  tasks:
    - name: "master"
      replicas: 1
      minAvailable: 1
      spec:
        containers:
          - command: ["sh","-c","horovodrun -np 2 --min-np 2 --max-np 5 --host-discovery-script discover_hosts.sh ..."]
    - name: "worker"
      replicas: 1
      spec:
        containers:
          - command: ["sh","-c","service ssh start && tail -f /dev/stdout"]
```

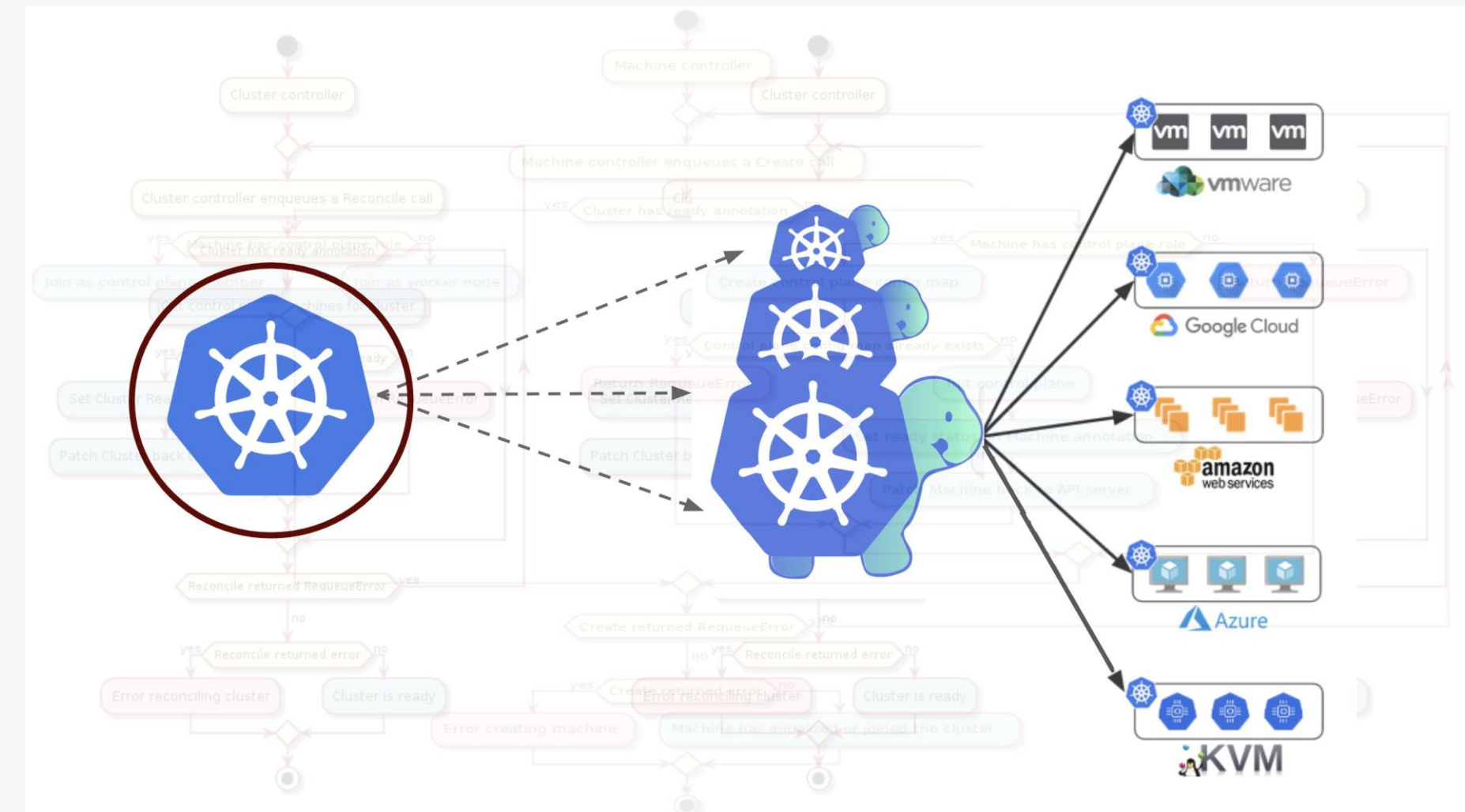
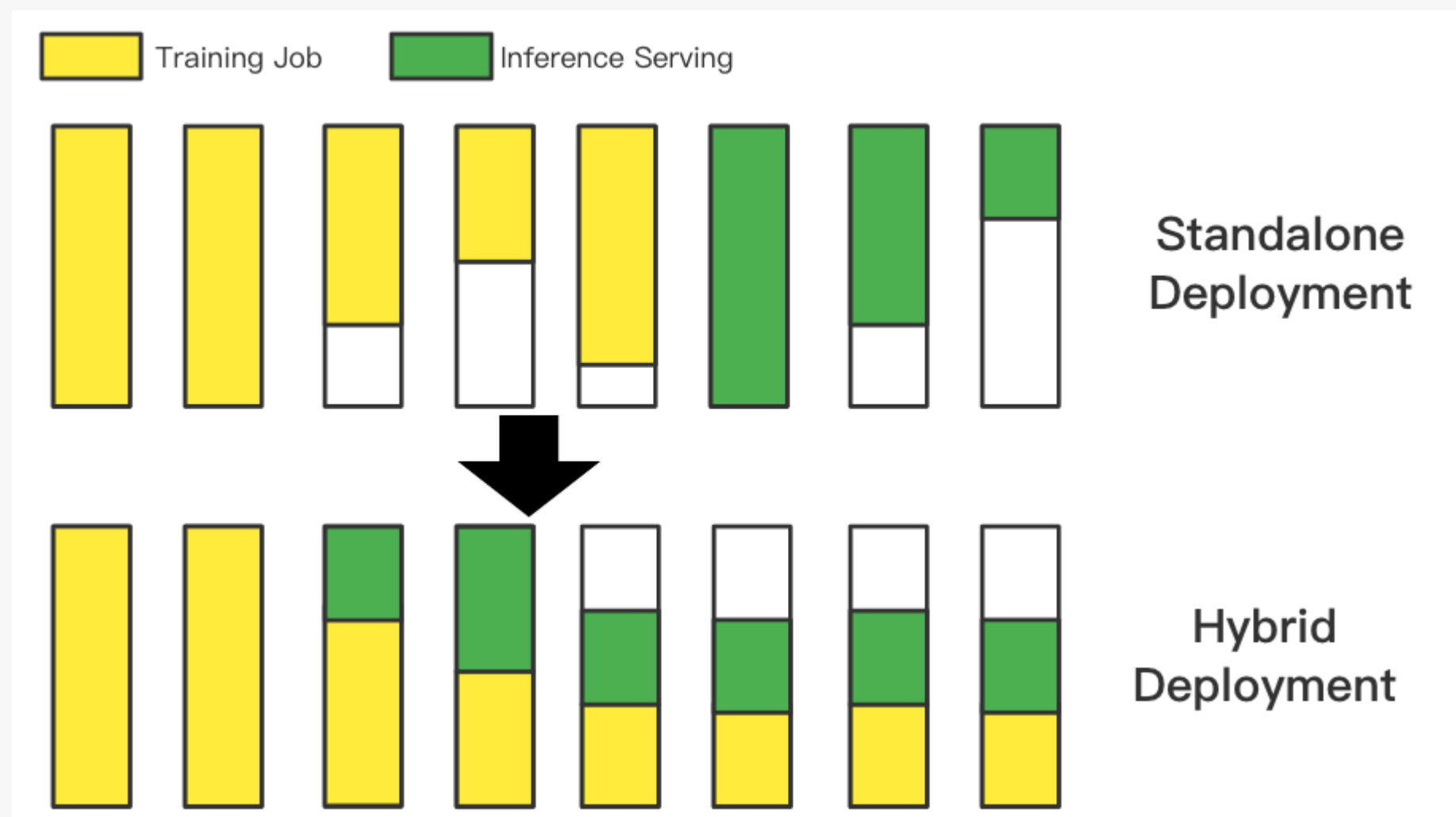


04

The Future



Future Focuses



- 训练作业和推理服务混合部署
- GPU共享
- 作业跨集群迁移

Join Volcano



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



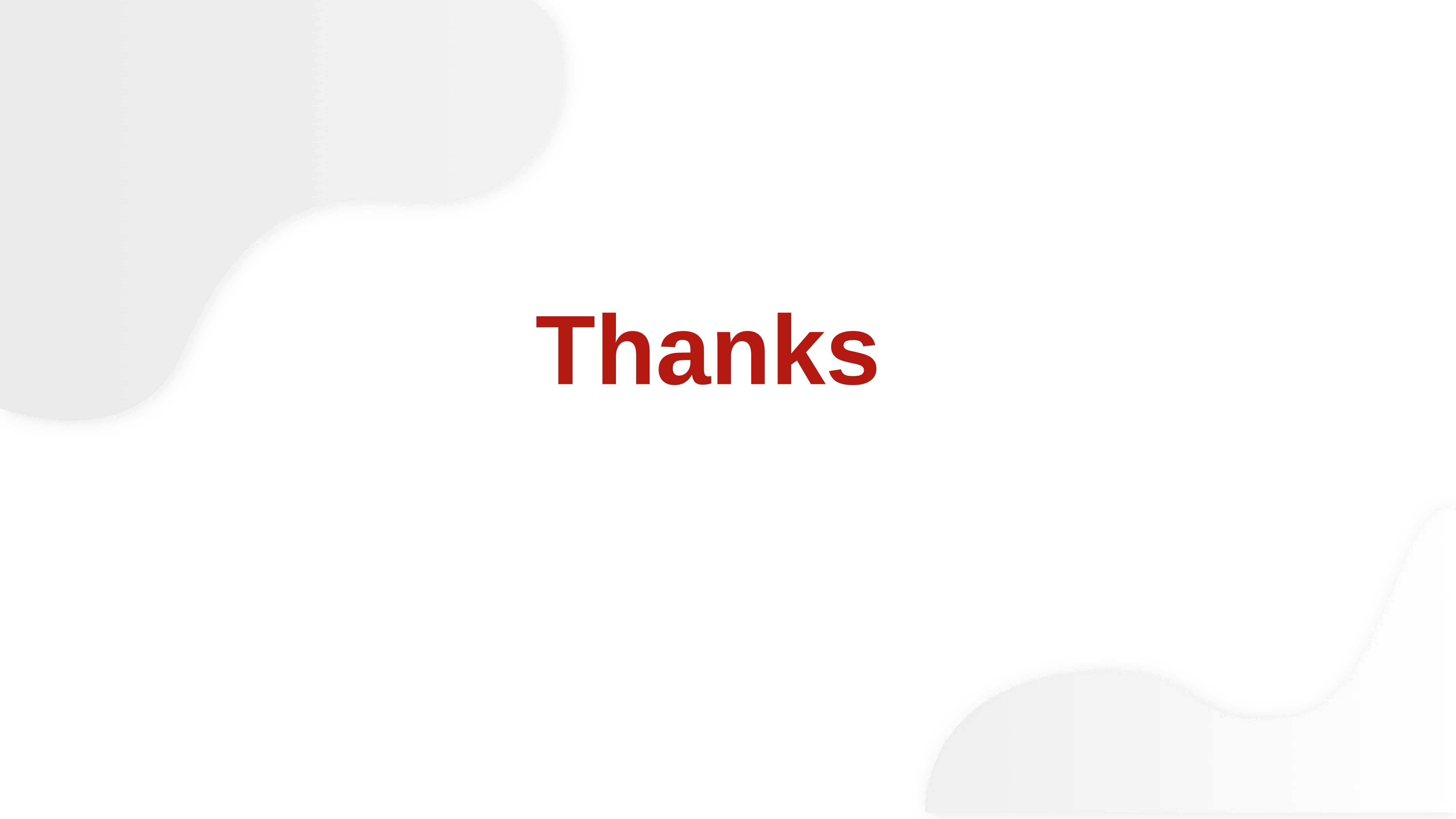
Slack Channel: <https://volcano-sh.slack.com/>



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Join Group



Thanks